# Venkata Lakshmi Parimala Pasupuleti

Sr Analyst Model Risk & AI · GenAI Model Validation · Agentic AI · LangGraph · RAG · LLM Evaluation · Python · AWS

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## PROFESSIONAL SUMMARY

AI/ML engineer with 7+ years production experience, now specializing in **GenAI and agentic AI model validation** at Intuit — building LLM evaluation frameworks (LLM-as-judge + golden datasets) to validate financial analytics AI before production, with structured audit trails grounding model outputs. Built **AgentProbe**, an open-source GenAI evaluation toolkit scoring outputs from 5 LLM providers on factual accuracy, hallucination rate, retrieval quality, completeness, and compliance — exactly the **non-deterministic AI system testing** model risk requires. **Leasel** is an 8-agent LangGraph compliance platform: prompt engineering, context engineering, hybrid-search RAG, HITL escalation, and LLM-as-judge evaluation achieving **93.6% precision, 87.9% F1** across 1,240 test cases. Deep stack: Python, SQL, LangGraph, LangChain, HuggingFace, OpenAI APIs, GCP/Vertex AI, AWS. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

## TECHNICAL SKILLS

**GenAI & Model Risk:** GenAI Model Validation, Agentic AI Testing, LLM-as-Judge, LLM Evaluation Frameworks, Non-Deterministic AI Testing, Challenger Solutions, Prompt Engineering, Context Engineering, RAG, Hallucination Detection, Model Risk Governance, Model Documentation & Audit Trails

**Python & ML:** Python (9+ yrs), LangGraph, LangChain, FastAPI, scikit-learn, XGBoost, PyTorch, Pandas, NumPy, HuggingFace, spaCy, BERT, ChromaDB, Pinecone

**Cloud & AI Tools:** GCP/Vertex AI, AWS (S3, EC2, RDS, SageMaker, Glue), Azure OpenAI, OpenAI APIs, Groq, Ollama, Docker, GitHub Actions CI/CD, Git

**Data & Analytics:** SQL, Snowflake, PostgreSQL, SQLite, ETL Pipelines, Data Quality Monitoring, Statistical Modeling, Regression Analysis

## WORK EXPERIENCE

### Intuit

Jun 2026 – Present

**Software Engineer — GenAI Validation & Model Risk · Financial Analytics**

- **Build and execute GenAI model validation frameworks** for financial analytics AI: LLM-as-judge evaluation suites + golden datasets validating model conceptual soundness, detecting hallucination and quality regressions before production; design **challenger solutions for non-deterministic AI systems** — variability analysis, confidence interval testing, and statistical evaluation of LLM/agentic outputs; compile model validation reports with structured audit trails and findings for stakeholder review.
- Integrate Anthropic Claude into Python/FastAPI services for financial data summaries and natural-language queries; add **model risk guardrails** (input validation, sensitive-data handling, output fallbacks); collaborate with data science teams to **maintain model risk governance documentation** and align validation practices with internal AI governance standards.
- Build end-to-end ML pipelines (scikit-learn, XGBoost) on **Snowflake** and **AWS** from feature engineering through model evaluation and prediction APIs; ship via GitHub Actions CI/CD with logging and monitoring for production visibility.

### Entrata — Multifamily & Affordable Housing SaaS

Jan 2017 – Jul 2024

**Analytic Software Engineer — Data Quality & Compliance Analytics**

- Automated compliance analytics pipelines across 1M+ financial records (income certifications, LIHTC/HCV/HUD data) — built data quality monitoring tools flagging inconsistencies before downstream reporting; maintained **100% PHI compliance, zero violations**; managed security mechanisms and access governance via AWS S3 Libsodium-encrypted pipelines with role-based controls.
- Optimized **200+ SQL queries (30% performance improvement)**; Python/Bash automation eliminating **80% of manual workload**; Docker containerization reducing microservices execution time 50%; delivered across **120+ Agile sprints**.

## PROJECTS

### AgentProbe — Open-Source GenAI Evaluation & Model Risk Toolkit

GenAI Validation · LLM Eval

Python · FastAPI · LangGraph · ChromaDB · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama — github.com/venkatalakshmiparimala/AgentProbe

- Built a **5-metric GenAI evaluation engine** (factual accuracy, hallucination rate, retrieval quality, completeness, compliance) routing prompts across 5 LLM providers via 3 LangGraph agents; **challenger solution framework** with CI golden-dataset regression tests; 15-endpoint FastAPI backend; variability analysis and outcome testing across non-deterministic AI outputs.

### Leasel — 8-Agent LangGraph Agentic AI Validation Platform

Agentic AI · RAG · LLM-as-Judge

Python · LangGraph (8-agent) · FastAPI · Hybrid RAG (TF-IDF + semantic) · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Designed and built an 8-agent LangGraph StateGraph (triage → retrieval → reasoning → validation → HITL escalation) with **prompt engineering, context engineering**, and hybrid-search RAG over 1,240 lease PDFs; **LLM-as-judge CI evaluation: 93.6% precision, 87.9% F1**; every determination traceable via full metadata and audit trail; core rule: LLM reasons, deterministic logic makes every regulated call.

### CrashClarity — Predictive Analytics on 104K Records

ML · Statistical Modeling

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages — Live Dashboard

- Random Forest + Logistic Regression ensemble on 104K+ UK accident records; feature engineering and **statistical diagnostic tests** achieving **85%+ classification accuracy**; deployed self-service BI dashboard translating model outputs to strategic insights.

## EDUCATION

**M.S. Data Analytics Engineering** — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Statistical modeling, ML pipelines, LLM evaluation, anomaly detection, data governance

**B.Tech Computer Science & Engineering** — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017